



PROGRAMMING FUNDAMENTALS.

C++ I.

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/*=====
DOCUMENT 1 – PRACTICE VERSION
C++ LEVEL 1 INTERVIEW / ASSESSMENT – INTERVIEW III (PRACTICE)
=====*/
```

```
/*
OVERVIEW
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```

This is a practice assessment designed to help you prepare.

Each question requires:

- (1) a written explanation
- (2) a FUNCTION that solves the problem

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*/
```

```
/*=====
REQUIRED FILE NAME
=====*/
```

Submit ONLY ONE file named:

interviewIII.cpp

```
/*=====
REQUIRED STRUCTURE (MANDATORY ORDER)
=====*/
```

```
/*
FOR EACH QUESTION:
```

```
/*
QUESTION X:
<FULL question>
```

```
ANSWER:
<Explanation>
*/
```

```
<FUNCTION PROTOTYPE>
```

AFTER ALL QUESTIONS:

```
int main() {
    return 0;
}
```

AFTER main():

ALL FUNCTION DEFINITIONS
*/

/*=====

PRACTICE QUESTIONS

=====*/

/*

QUESTION 1

Write a function that receives an integer and returns:

1 if positive, -1 if negative, 0 if zero.

Also print the classification.

*/

/*

QUESTION 2

Create a function that receives three integers and returns the largest.

Explain how it generalizes the 2-number case.

*/

/*

QUESTION 3

Write a function that receives an array and its size and returns BOTH:

- sum
- average (use reference parameters)

Explain why references are useful.

*/

/*

QUESTION 4

Write a function that swaps two integers using pointers.

Explain step-by-step how memory is modified.

*/

/*

QUESTION 5

Define a struct Student.

Write a function that receives a Student and determines if the student is passing (grade $\geq$ 70).

*/

/*

QUESTION 6 - C-STRINGS

Write a function that counts characters in a C-string WITHOUT strlen().

Also count only alphabetic characters.

*/

```

/*
QUESTION 7 - C-STRINGS
Write a function that reverses a C-string in place (no extra array).
*/

/*
QUESTION 8 - STRING CLASS
Write a function that counts vowels AND consonants in a string.
*/

/*
QUESTION 9 - VECTORS
Write a function that returns BOTH:
    - largest value
    - index of largest value
*/

/*
QUESTION 10 - STRUCT + VECTOR
Define struct Course.
Write a function that returns total credits from a vector of Course.
*/

/*=====
    RULES (APPLY STRICTLY)
=====*/

/*
- Each question MUST have a function
- Prototype MUST be under the question
- DO NOT group prototypes at the top
- DO NOT call functions in main()
- Code MUST compile
*/

/*=====
    END PRACTICE VERSION
=====*/
Any questions, email me.
LDV.

```